

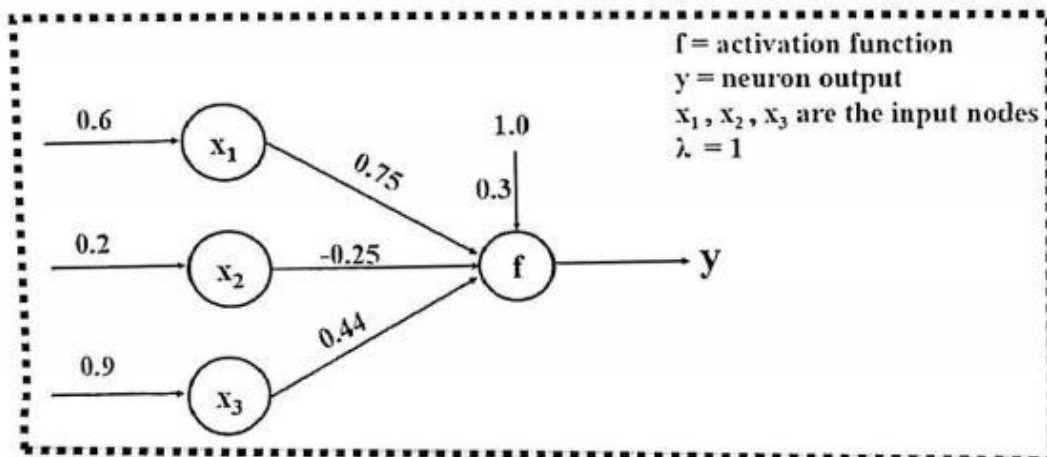
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B.E - (Computer) (Semester-VII)(Revised Course 2019-2020)**CEAM 04 Neural Networks and Deep Learning****[Time: 3:00 Hours]****[Max. Marks:100]**

- Instructions:**
1. Attempt two questions from Part-A, two questions from Part-B and one from Part-C.
 2. Figures to the right indicate marks.
 3. Make suitable assumptions wherever necessary.

PART-A

- Q1**
- a) Draw a neat sketch and explain Biological Neuron Model. **06**
 - b) Explain Perceptron Learning Rule with the help of an example. **06**
 - c) Refer to Figure 1 and compute the Output for the following activation functions: **08**
 - i) Linear Activation Function
 - ii) Logistic Sigmoid Activation Function
 - iii) Bipolar Sigmoid Activation Function
 - iv) Hyperbolic Tangent Activation Function

**Figure 1**

- Q2**
- a) What are the different methods to choose number of hidden layers neurons and learning constant? **07**
 - b) Given the training set as (1, 1, 1), (-1, -1, -1), (-1, -1, 1), design Brain-State-in-a-Box (BSB) network. Test network response for corrupted input (-0.2, -0.5, -0.8). **07**

- c) Explain advantages and disadvantages of ART Neural Network. **06**

- Q3** a) Describe Error Backpropagation Algorithm for training a artificial neural network. **08**

- b) Find if the prototype points presented below are linearly separable. If yes, provide the line equation in a 2D plane. **06**

Class I			
X1	X2	X3	X4
[5,2]	[6,2]	[4,3]	[5,5]
Class II			
X5	X6	X7	X8
[0,0]	[-2,-1]	[-2,4]	[-4,0]

- c) Explain Reinforcement Learning. **06**

PART-B

- Q4** a) Explain the following: **10**

- i) Necessity of Deep Neural Networks
ii) Structured Outputs in Convolution Neural Network

- b) Elaborate in detail the convolution operation used in CNN. **10**

- Q5** a) Explain the following: **10**

- i) Sparse Autoencoder
ii) Denoising Autoencoder

- b) Explain Long Short-Term Memory Model with an appropriate block diagram. Also mention its advantages and disadvantages. **10**

- Q6** a) Elaborate on Data Types in Convolution Neural Network. **05**

- b) What are Deep Belief networks? **05**

- c) Explain the following: **10**

- i) Bi-directional Recurrent Neural Network
ii) How does having separable kernel make the Convolution Neural Network efficient?

PART-C

- Q7**
- a) Explain any two applications of Artificial Neural Networks. **05**
 - b) Differentiate between Feed Forward Neural Networks and Recurrent Neural Networks. **05**
 - c) What is an undercomplete autoencoder? **05**
 - d) What are Self Organizing Maps? **05**
- Q8**
- a) Explain Simple Competitive Learning Concept. **05**
 - b) Describe Widrow-Hoff Learning Rule. **05**
 - c) Elaborate Padding Strategies in Convolution Neural Network. **05**
 - d) Write a short note on Boltzmann Machine. **05**